



Sri Lanka Institute of Information Technology

Procurement for Construction Industry

Critical Reflection Report

Case Studies in Software Engineering SE3070

Group ID: SE3070_WE_83

Group Members:

1. IT20132682 - Abilash.L
2. IT20051648 - Danushan.R
3. IT20072742 - Agash.V
4. IT20250874 - T.Abinaya

(All members contributed towards all the diagrams as well as the writing of the report)

Introduction

The procurement management system for the construction industry is to deal with the difficulties faced with procurement procedures within the construction industry due to its inherently distributed environment. Here we aim to design a software system to manage procurement processes in the construction industry.

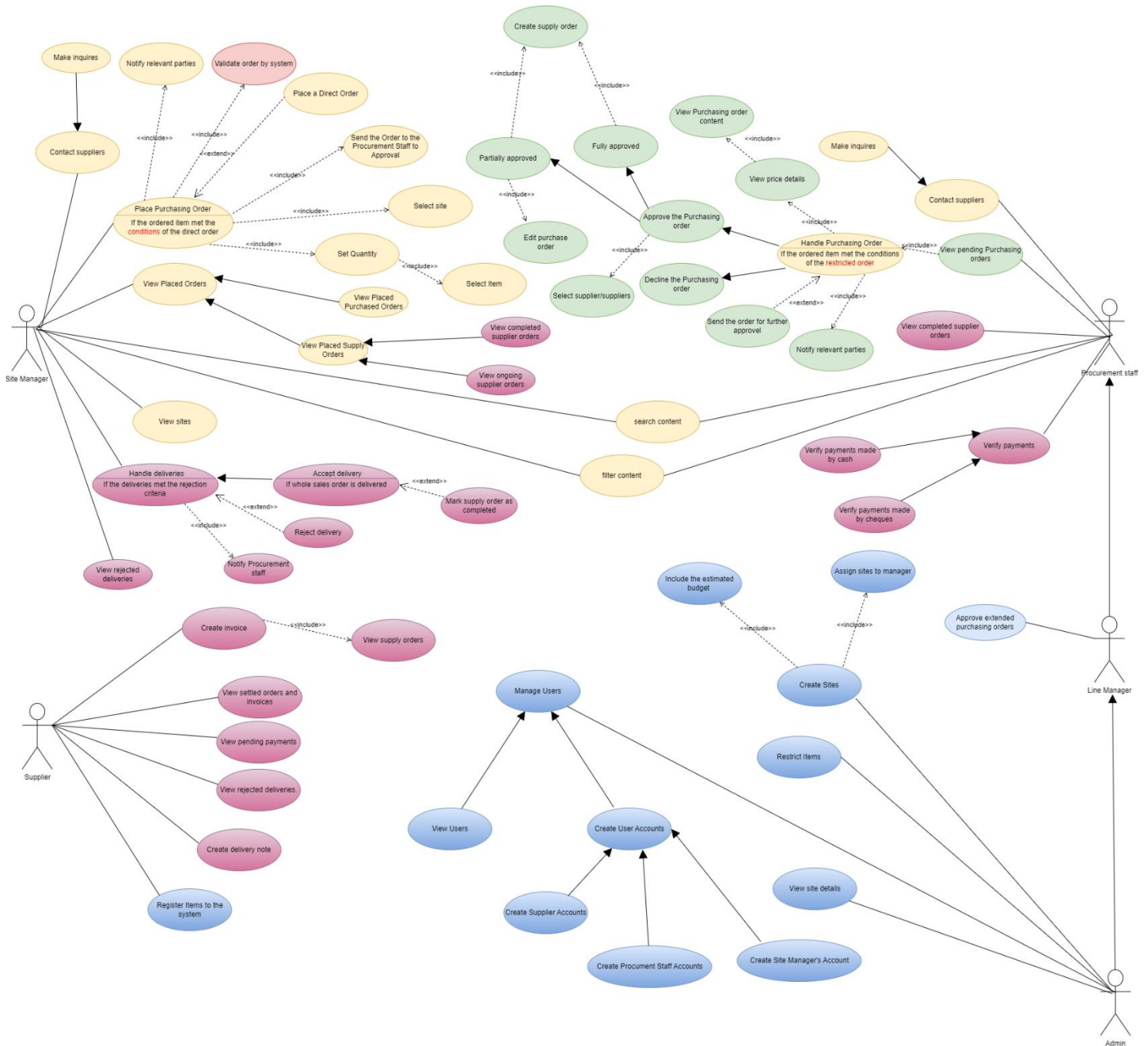
Furthermore, the system generates, and places purchase orders, makes inquiries about previously placed orders and the goods delivered to return will be handled by the system. This system is basically based on four main users as site manager, procurement staff, manager and suppliers.

Actors in this system:

- Site manager
- Procurement staff member
- Line Manager
- Administrator
- Supplier

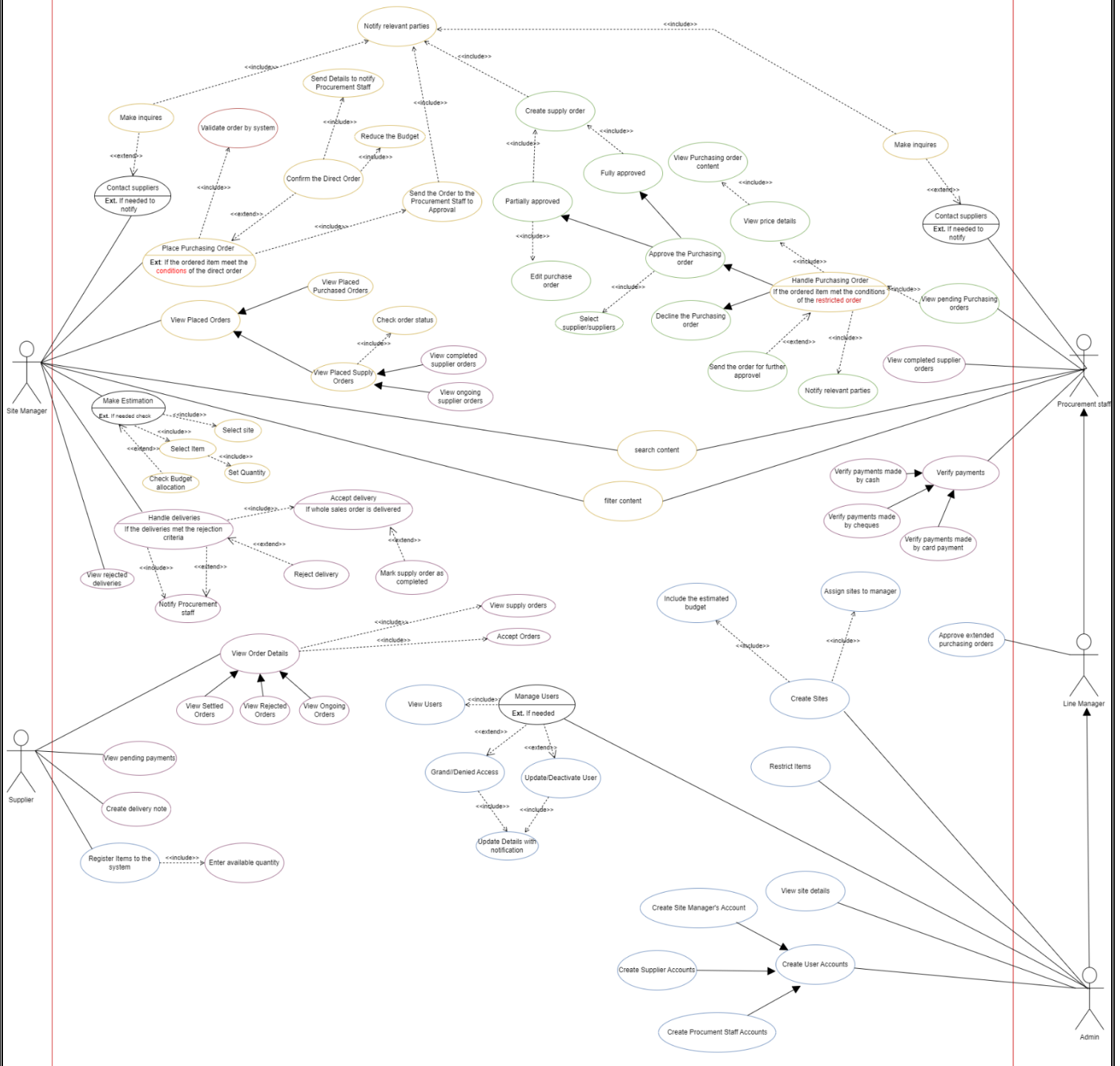
1. Use Case Diagram

1.1 Given Use Case Diagram



1. 2 Updated Use Case Diagram

Procurement for Construction Industry



1.3 Modifications & Justifications

1. Modification: Add include use case to the “Confirm the Direct Order” use case.

Justification: Because if the order is confirmed, procurement staff should be notified to check the order and the budget will be reduced for the future orders.

2. Modification: Create a new main use case to “Make Estimation”.

Justification: Site Manager can do some estimation with the help of allocated budget and the site details and the item details.

3. Modification: Make some changes about the notify the persons.

Justification: While making the inquiries, the relevant parties will be notified.

4. Modification: Add generalization for the View order details for the Supplier actor.

Justification: Supplier can view the orders like settled orders, Rejected orders, ongoing orders.

5. Modification: Add include use case while supplier register the items.

Justification: Instead of the item’s details only, the quantity also would be added. It will reduce the conflict between the site manager and the supplier.

6. Modification: Do some modifications in the manage user use case of admin.

Justification: While admin manage the users, first need to view the users and then if admin wants to deactivate the user or maybe admin can grant or denied some permissions of the user.

7. Modification: Added include use cases for Supplier View orders

Justification: While view the order details, suppliers can accept or denied the orders and view the supply orders too.

8. Modification: Add some new payment methods.

Justification: Card payment is mandatory while making the payments.

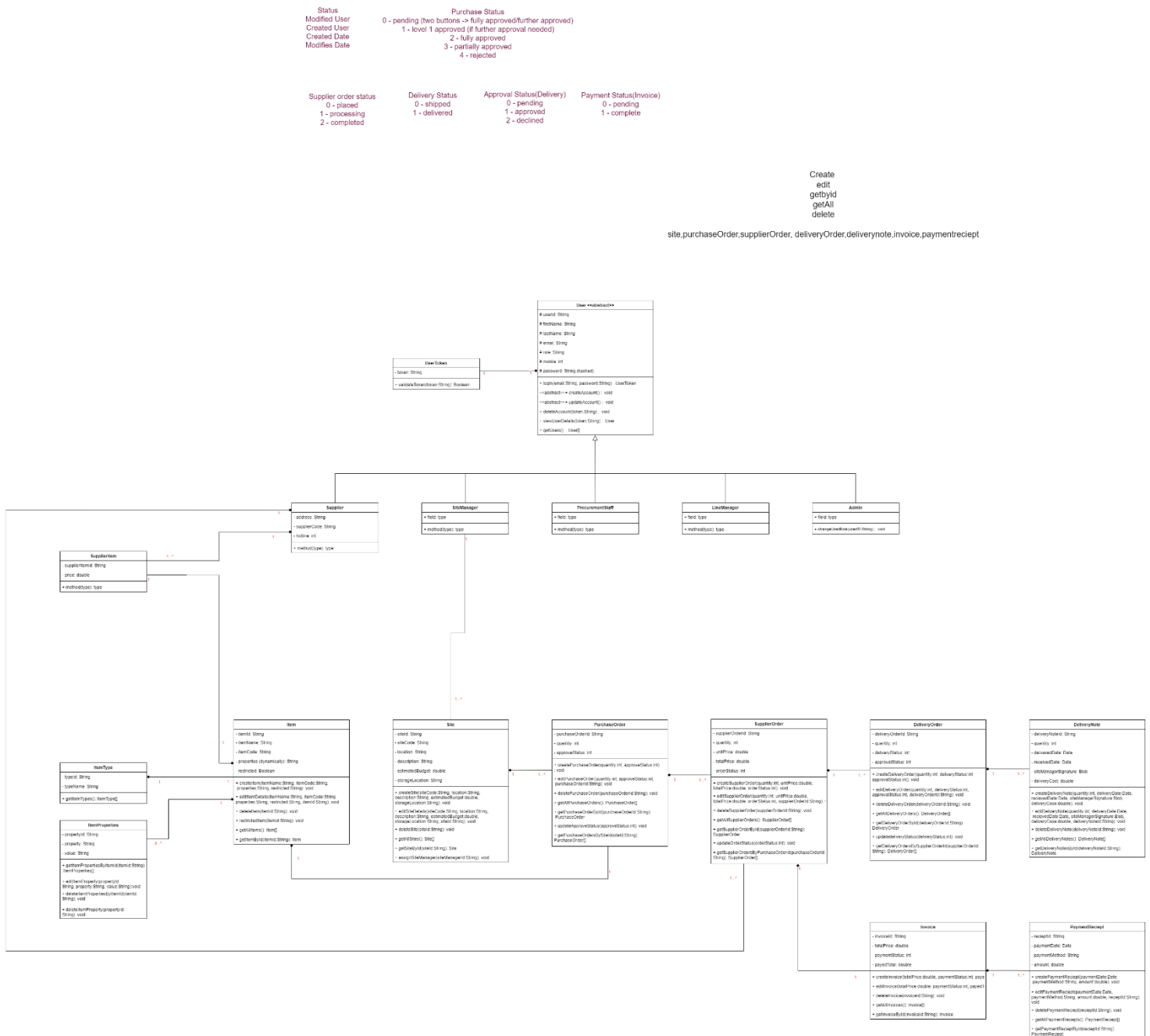
9. Modification: Add some changes for the partially approved orders in procurement staff.

Justification: While partially approve the orders, the quantity can be changed.

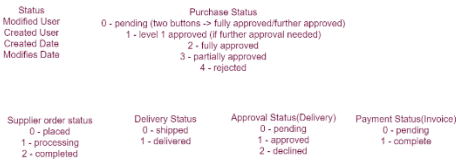
10. Modification: Add include for create supply orders in Procurement staff.

Justification: While creating the orders, the relevant parties should be notified.

2.1 Given Class Diagram

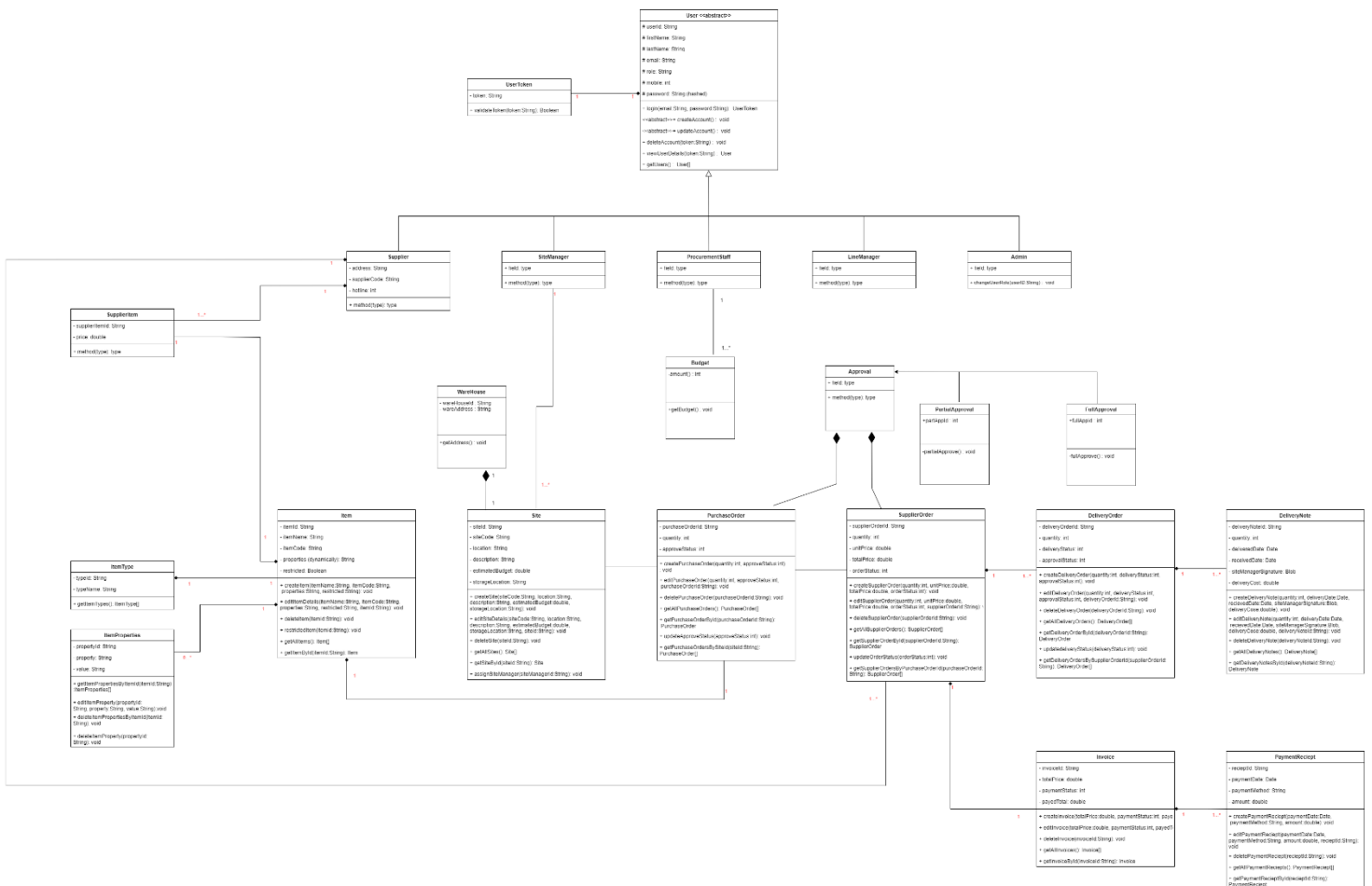


2.2 Updated Class Diagram



Create
edit
getbyid
getAll
delete

site,purchaseOrder,supplierOrder, deliveryOrder,deliverynote,invoice,paymentreceipt



1.3 Modifications & Justifications

1. Modification: Add Approval class.

Justification: In order to maintain the approval details and status of the approval, we need this class.

2. Modification: Add two child classes to the Approval classes.

Justification: There can be two types of approval, one is Partially approved and another one is fully approved.

3. Modification: Add relationships between the Supplier Orders, Purchase Orders with Approvals.

Justification: Approvals cannot exist without Supplier Orders, Purchase Orders.

4. Modification: Change the relationship between Supplier Orders, Purchase Orders.

Justification: Instead of composition, added association, because of the loosely coupling.

5. Modification: Add new class called Warehouse.

Justification: Site has warehouse to store the orders. So, need to consider the capacity.

6. Modification: Did some changes in cardinality.

Justification: According to the requirements, some changes have been done in the cardinality.

7. Modification: The relationship between the Purchase Orders and site was changed.

Justification: Instead of composition, added association, because of the loosely

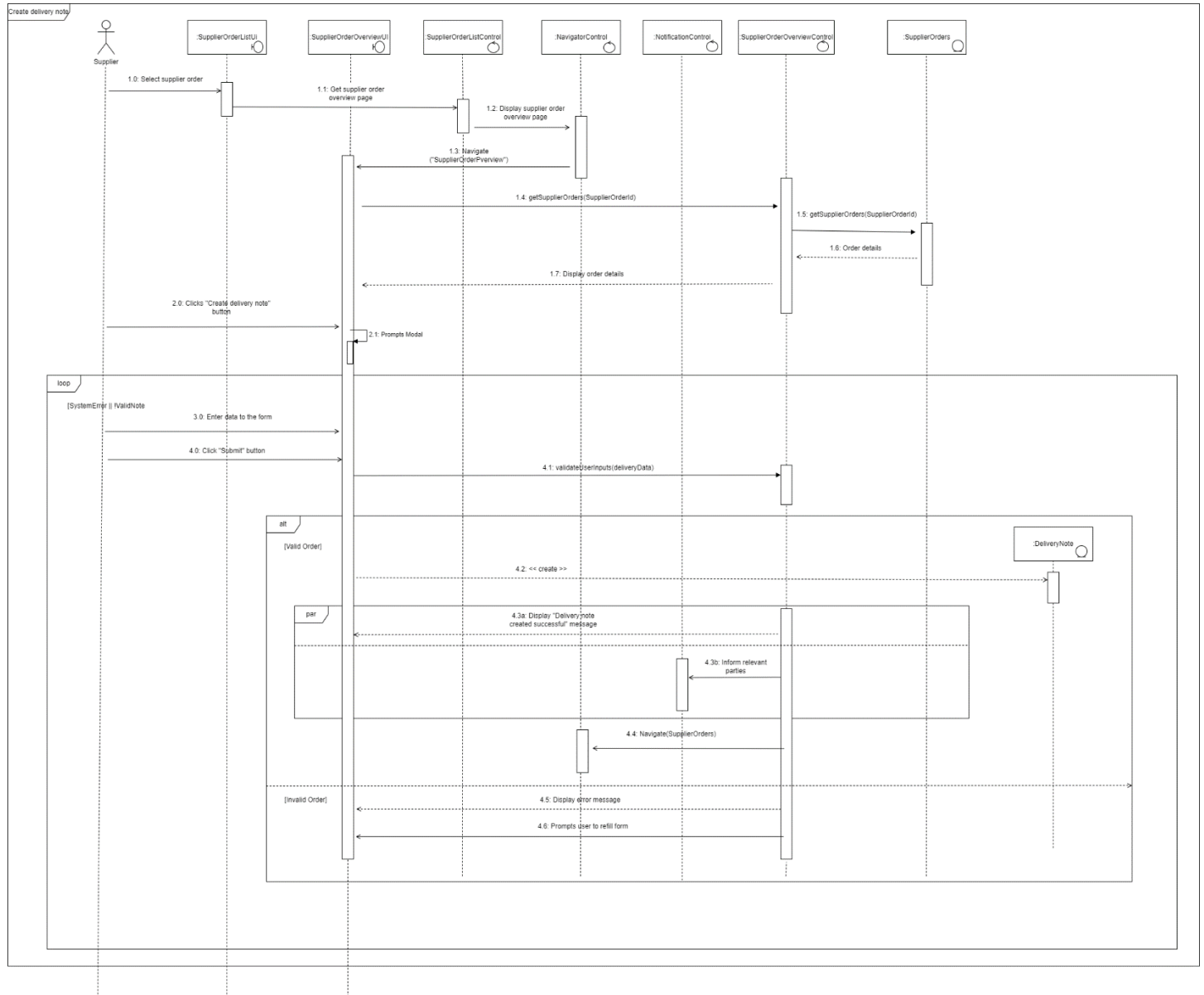
8. Modification: Add some new payment methods.

Justification: Card payment is mandatory while making the payments.

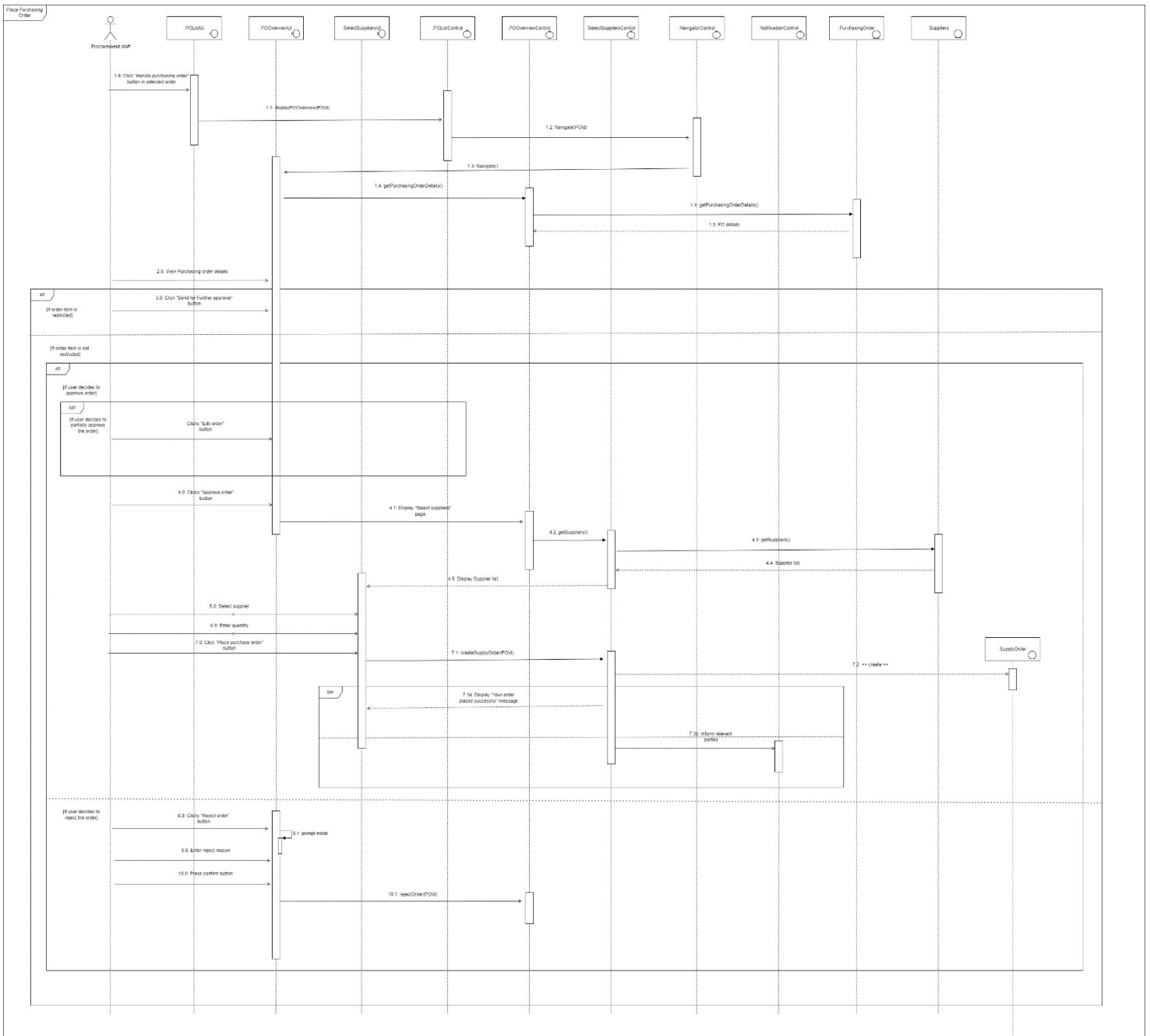
3. Sequence Diagram

3.1 Given Sequence Diagrams

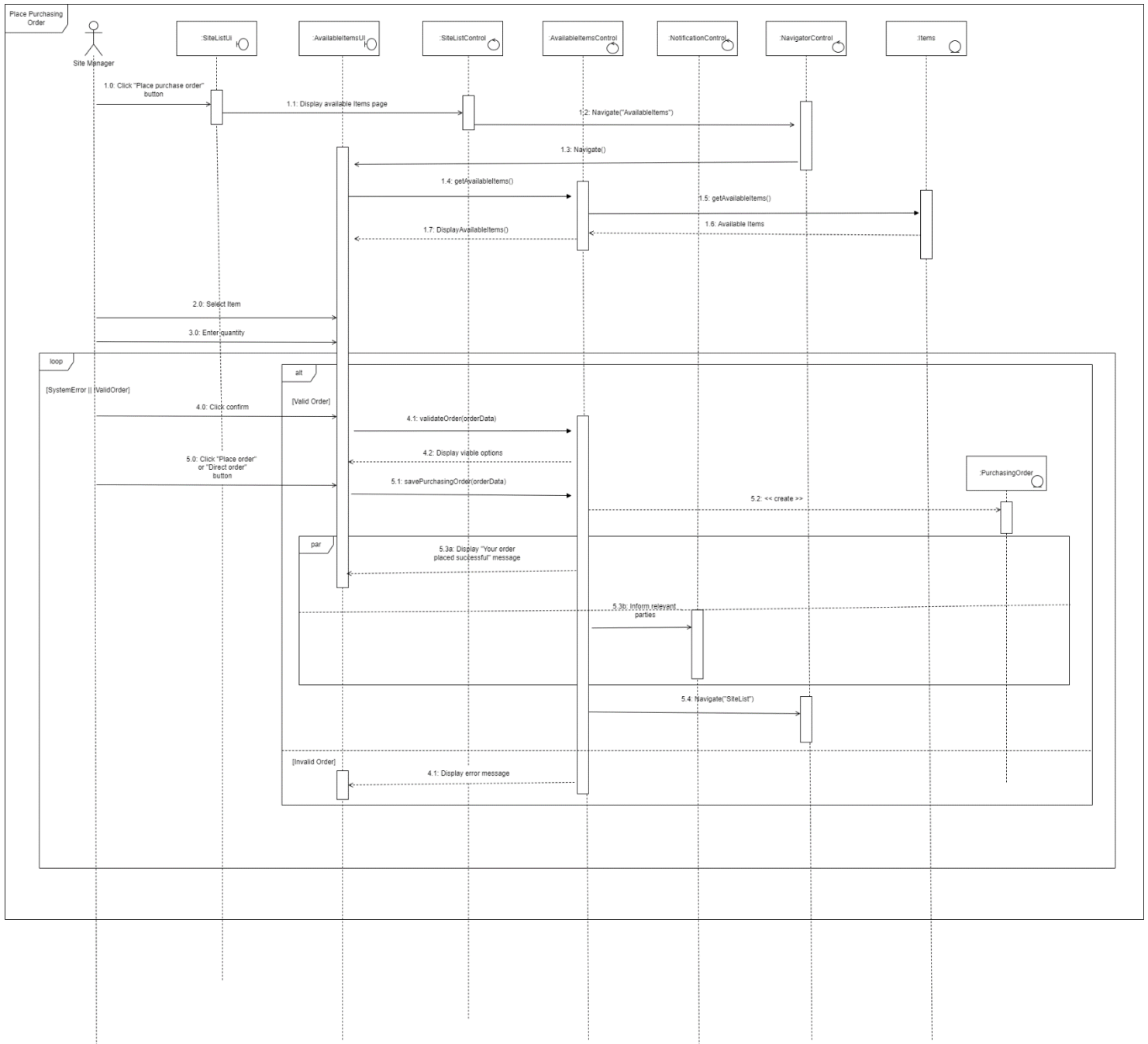
a. Create Delivery Note



b. Handle Purchasing Order

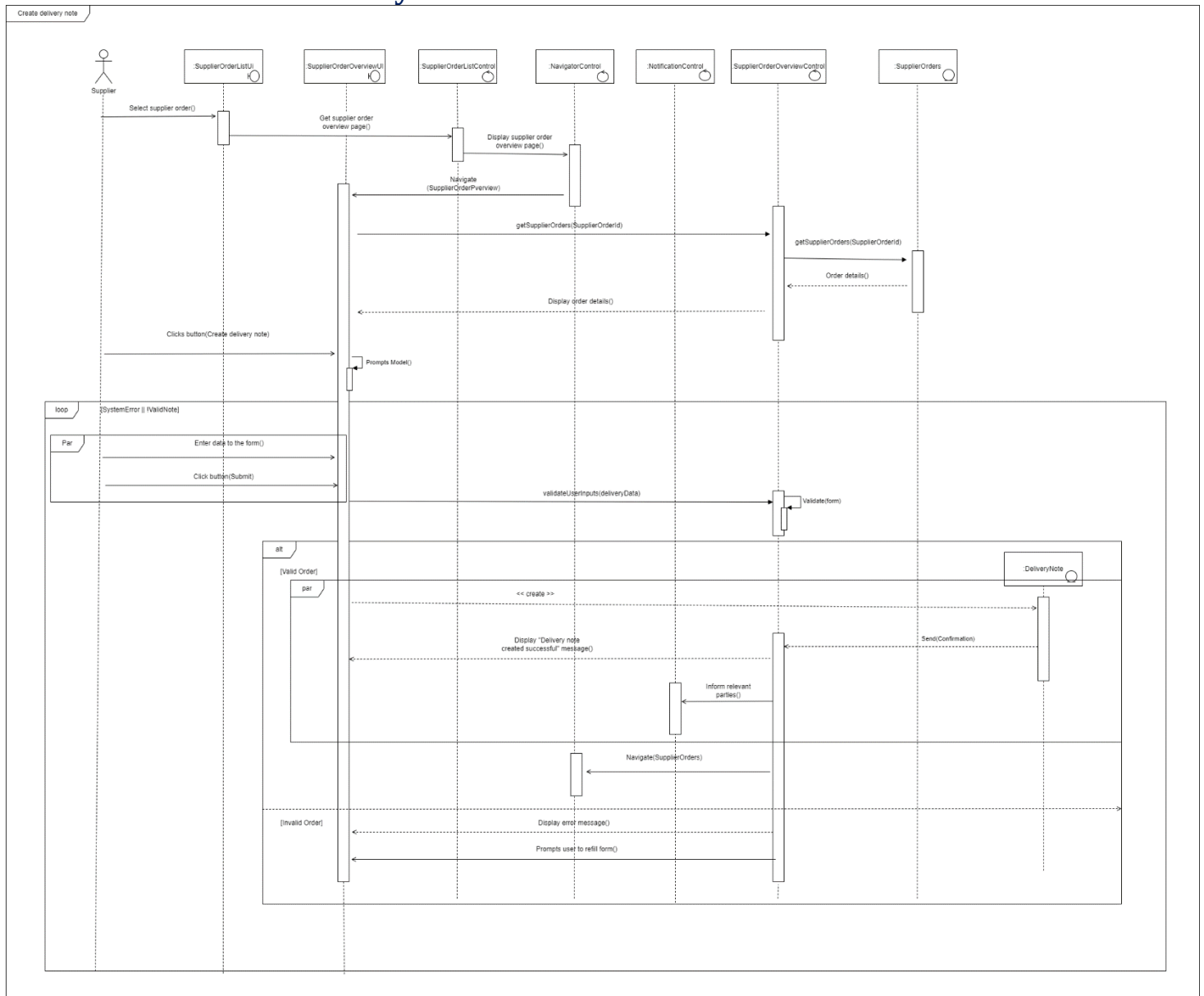


c. Place Purchasing Order

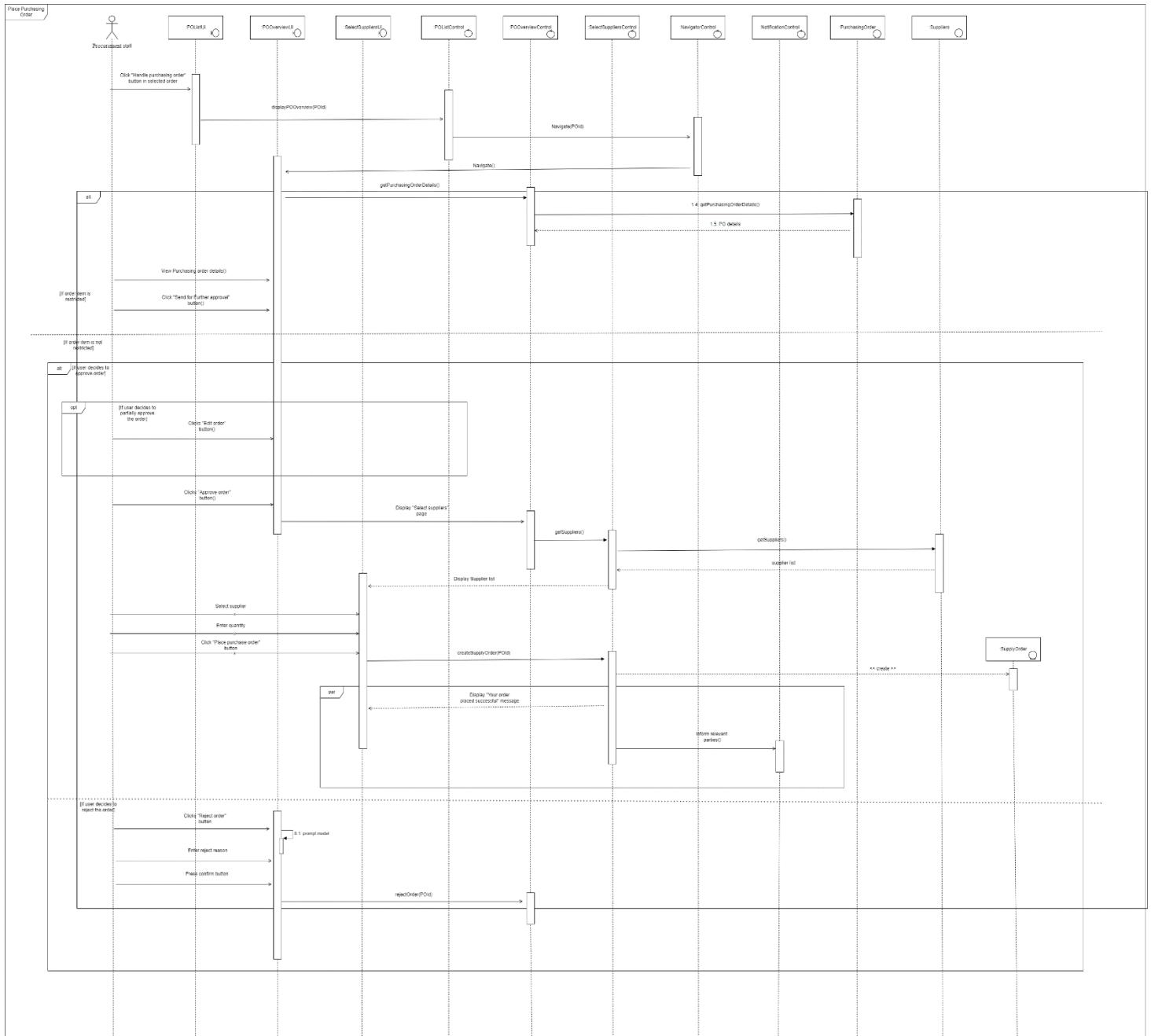


3.2 Updated Sequence Diagrams

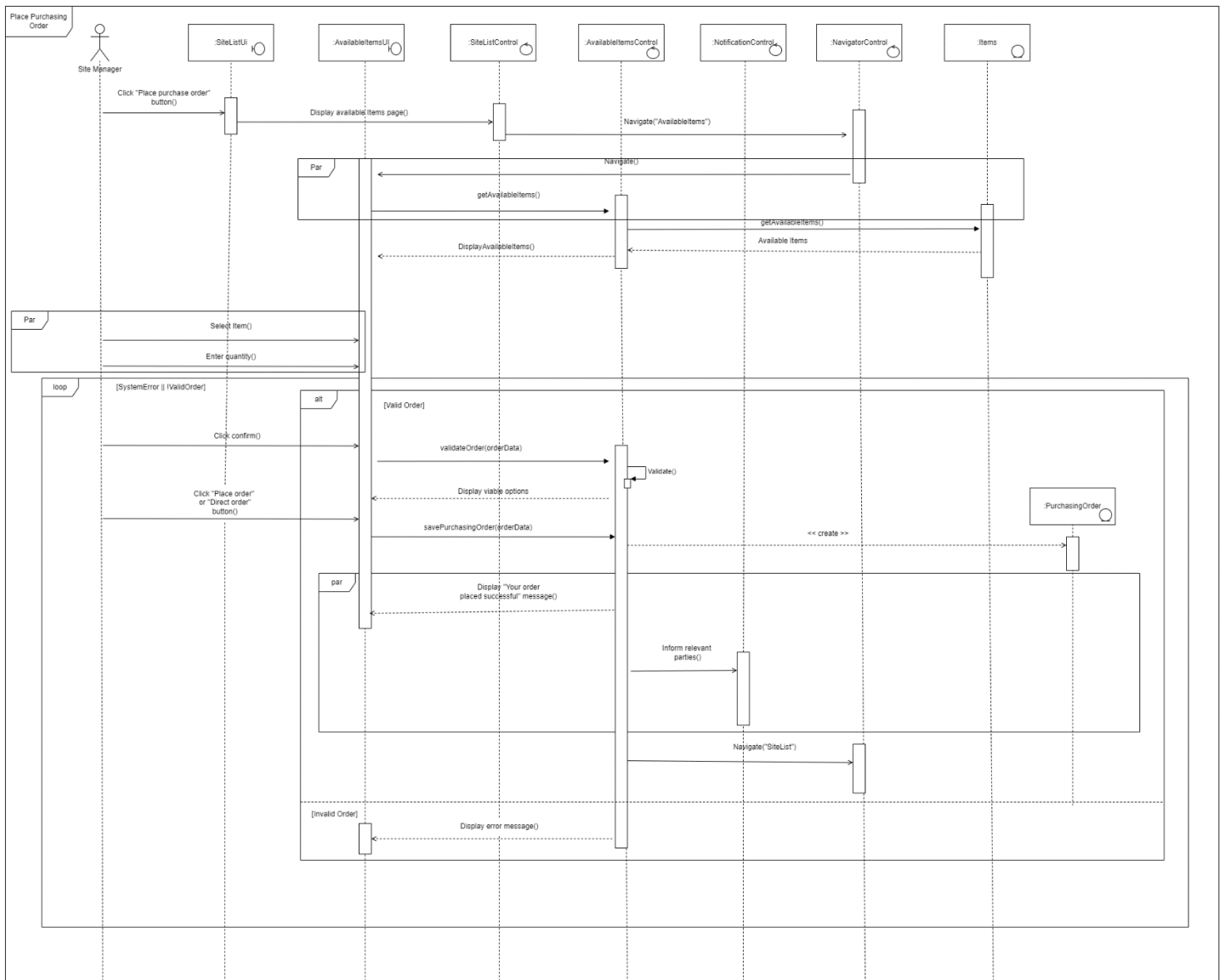
a. Create Delivery Note



b. Handle Purchasing Order



c. Place Purchasing Order



3.3 Modifications & Justifications

1. Modification: In Create Delivery Note, while submitting the form, a parallel tag was added.

Justification: Both filling and submitted is done in User Interface, so it can be considered as a parallel process.

2. Modification: After the form submission, the self-call was added in the controller.

Justification: In order to validate the form, this self-call should be there.

3. Modification: The number was removed.

Justification: The numbering the process is for communication diagram. In sequence diagram, it is not compulsory.

4. Modification: In Place Purchasing Order while submitting the form, a parallel tag was added.

Justification: Both filling and submitted is done in User Interface, so it can be considered as a parallel process.

5. Modification: After the form submission, the self-call was added in the controller.

Justification: In order to validate the form, this self-call should be there.

6. Modification: Added destroy for the created object.

Justification: If the object is created in middle of the process, it should be destroyed after its usage is done.

7. Modification: Did some changes in message calls.

Justification: Instead of asynchronous message, the synchronous message was added to continue the communication between the system and the users.

8. Modification: In Handle Purchasing orders some parallel activities were added.

Justification: Instead of sequential, parallel tag was added to make the flow easily.

4. UI Design

4.1 Given UI Designs

4.1.a Mobile App Designs



4.1.b Web App Designs

View All Purchase Orders Page

← Purchase Orders

Show 10 entries

Search:

No	Site Code	Item Code	Status	Created Date	Last Updated Date	Actions
1	TB20568	RM10256	Fully Approved	20/09/2020, 03:00:00	22/09/2020, 03:00:00	
2	TB20568	RM10256	Fully Approved	20/09/2020, 03:00:00	22/09/2020, 03:00:00	
3	TB20568	RM10256	Fully Approved	20/09/2020, 03:00:00	22/09/2020, 03:00:00	
4	TB20568	RM10256	Fully Approved	20/09/2020, 03:00:00	22/09/2020, 03:00:00	

Showing 1 to 10 of 1225 entries

Previous 1 2 3 Next

Detailed View of a Purchase Order

← Purchase Order Id

Site name

TUDAWE BROTHERS (PVT) LTD

Site Code

TB20568

Location

505/2,ELVITIGALA MAWATHA, COLOMBO 05

Description

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.

Storage Location

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.

Item Name

Tokyo Super

Type

Cement

Item Code

RM10256

Estimated Budgets (Approx)

LKR. 400,00.00

Requested Quantity

250

Further Approval Requirement

Get Order

Approve

Reject

View All Suppliers Page

← Select Suppliers

Place Order

Show 10 entries

Search:

Supplier Code	Address	Item Code	Quantity	Unit Price	Total Price	Select
SUP02856	No. 65/A, Galle Road, Colombo 03	RM10256	250	LKR. 3200.00	LKR. 800,000.00	☐
SUP02856	No. 65/A, Galle Road, Colombo 03	RM10256	250	LKR. 3200.00	LKR. 800,000.00	☐
SUP02856	No. 65/A, Galle Road, Colombo 03	RM10256	250	LKR. 3200.00	LKR. 800,000.00	☐
SUP02856	No. 65/A, Galle Road, Colombo 03	RM10256	250	LKR. 3200.00	LKR. 800,000.00	☐

Showing 1 to 10 of 1225 entries

Previous 1 2 3 Next

View All Supply Orders Page

← Supply Orders

Show 10 entries

Search:

No	Site Code	Item Code	Status	Created Date	Last Updated Date	Actions
1	TB20568	RM10256	processing	20/09/2022, 03:00:00	22/09/2022, 03:00:00	
2	TB20568	RM10256	processing	20/09/2022, 03:00:00	22/09/2022, 03:00:00	
3	TB20568	RM10256	processing	20/09/2022, 03:00:00	22/09/2022, 03:00:00	
4	TB20568	RM10256	processing	20/09/2022, 03:00:00	22/09/2022, 03:00:00	

Showing 1 to 10 of 1225 entries

Previous 1 2 3 Next

Detailed View of a Supply Order

← TB20568

Site name

TUDAWE BROTHERS (PVT) LTD

Site Code

TB20568

Location

505/2,ELVITIGALA MAWATHA, COLOMBO 05

Description

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.

Storage Location

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.

Order Status

Processing

Requested Quantity

250

Unit Price (LKR)

3200.00

Total Price (LKR)

800,000.00

Create Delivery Note (Modal)

Create Delivery Note

Quantity

Delivery Date

Received Date (Expect)

Delivery Cost (LKR)

Total Cost (LKR)

Submit

Cancel

Login Page

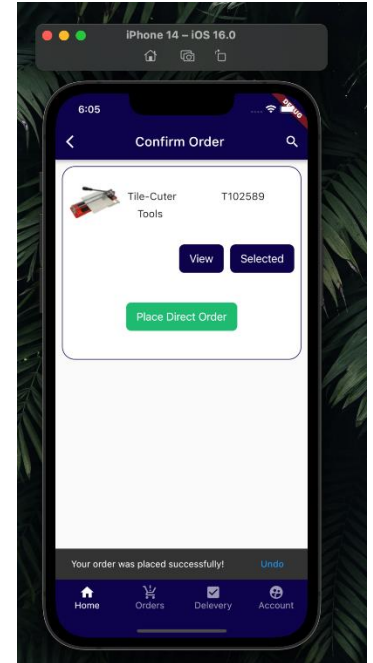
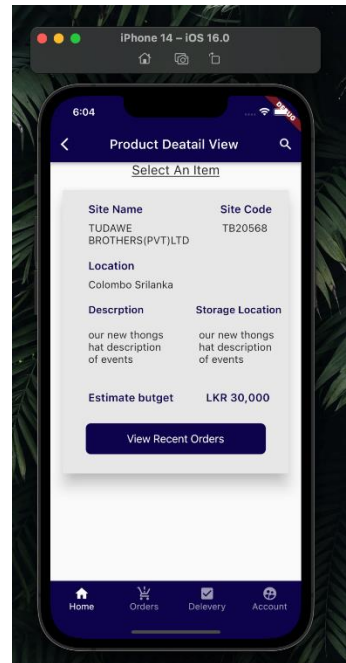
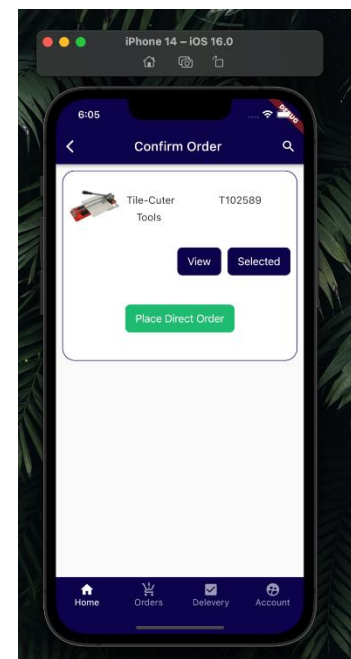
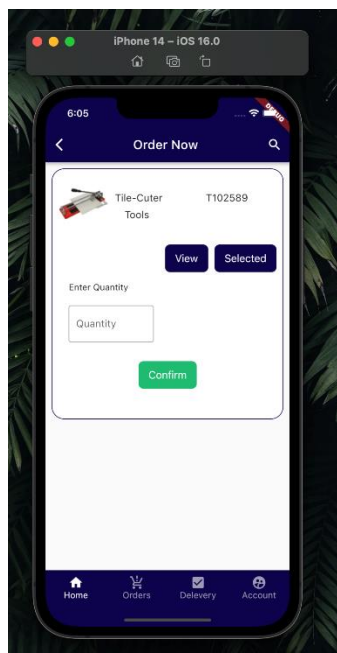
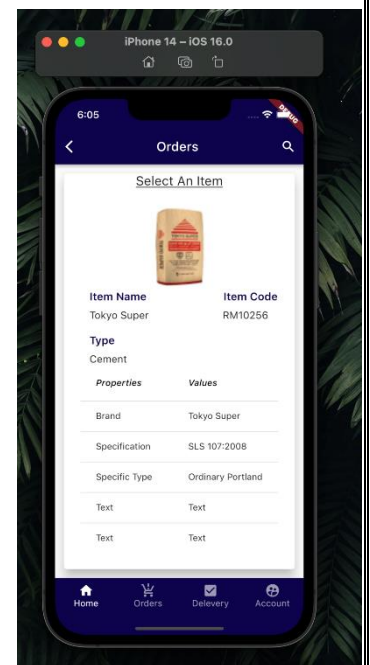
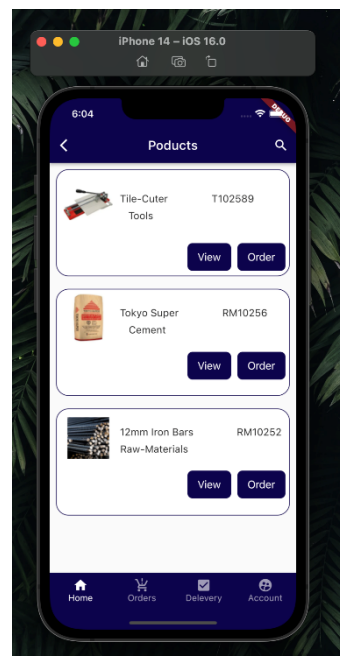
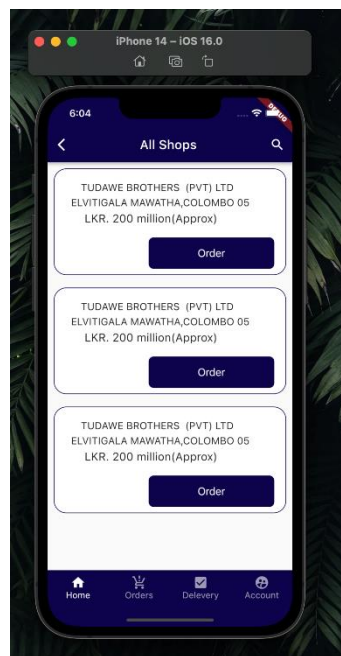
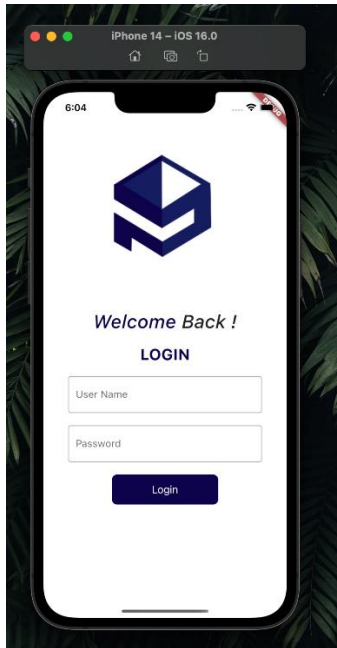


Welcome back!
Login

Login

4.2 Updated UI Designs

4.2.a Mobile App Designs



4.2.b Web App Designs

Sign in

Email address *

Password *

☐ Remember me

Sign in

[Forgot password?](#) [Don't have an account? Sign up](#)

Copyright © Your Website 2022

LOGO

PRODUCTS

PRICING

BLOG

R

Search

Country		Details		
Name	ISO Code	Population	Size (km²)	Density
India	IN	1,324,171,304	3,287,263	402.82
China	CN	1,403,500,365	9,596,961	146.24
Italy	IT	60,483,973	301,340	200.72
United States	US	327,167,434	9,533,520	33.27
Canada	CA	37,602,103	9,984,670	3.77
Australia	AU	25,475,400	7,692,024	3.31

Rows per page: 10 1-10 of 15

LOGO

PRODUCTS

PRICING

BLOG

R

Delivery Note

Quantity

Delivery Date

Received Date

Delivery Cost

Total Cost



Believing is seeing

Apple pro display XDR

Pro Display XDR\$5,999

Pro stand\$599

Vesa Mount Adapter\$199

Total\$7,197.00

EDIT ORDER

APPROVE ORDER

DECLINE ORDER

LOGO

PRODUCTS

PRICING

BLOG

R

Country		Details		
Name	ISO Code	Population	Size (km²)	Density
India	IN	1,324,171,304	3,287,263	402.82
China	CN	1,403,500,365	9,596,961	146.24
Italy	IT	60,483,973	301,340	200.72
United States	US	327,167,434	9,533,520	33.27
Canada	CA	37,602,103	9,984,670	3.77
Australia	AU	25,475,400	7,692,024	3.31

Rows per page: 10 1-10 of 15

4.3 Modifications & Justifications

Mobile App Designs, Web App Designs

1. Modification: Navigation bar was changed.

Justification: In the navigation bar they added more than 4 services to navigate. But it is not a user-friendly design, and it will lead to the complexity, and it will mess the application.

2. Modification: The list view of the products was changed.

Justification: List view is changed with modern card view. Because of the user-friendly and neatness.

3. Modification: Same sized of buttons used instead of different sizes.

Justification: Increase the attractive and user friendly.

4. Modification: Did some modification in the order page.

Justification: while order the goods, there was other goods details should not be displayed. So other details were removed.

5. Modification: Did some changes in the image size and font size, type.

Justification: For the attractiveness, the font and images should be in same size.

6. Modification: Header was added.

Justification: In order to navigate, there will be a header.

7. Modification: Some features were added in the login.

Justification: Reset password, view the password options were added to increase the user expectations.

8. Modification: Some complex textual formatting was removed.

Justification: If it is a complex text, it would be not good to read, So, some images were added to increase the user-friendly.